

Multiple Choice (10 marks)

1. The glomerular filtration rate is controlled primarily by alteration of the glomerular capillary pressure. What are the two factors which produce these changes?
 - (a) Glomerular capillary pressure and blood volume
 - (b) Diameter of the efferent arterioles and plasma protein concentration
 - (c) Renal blood flow and the concentration of sodium in the distal tubule
 - (d) Arterial blood pressure and diameter of the afferent arterioles
 - (e) Kidney size and blood urea nitrogen levels
2. What function does oxygen serve in the body?
 - (a) Oxygen is used to maintain the temperature of cells.
 - (b) Oxygen is used by cells to produce alcohol.
 - (c) Oxygen is used to produce carbon dioxide in cells.
 - (d) Oxygen is solely responsible for stimulating brain activity.
 - (e) Oxygen is used to synthesize proteins in cells.
3. About 0.5 percent of the American population have epilepsy, the falling sickness. Discuss the genetics of this disease.
 - (a) Epilepsy is the result of a chromosomal aneuploidy, similar to Down syndrome.
 - (b) Epilepsy is always inherited in an X-linked recessive pattern.
 - (c) Epilepsy is a dominant trait.
 - (d) Epilepsy is caused by an autosomal dominant trait.
 - (e) Epilepsy is due to an autosomal recessive trait.
4. Charles Darwin was the first person to propose
 - (a) that humans evolved from apes.
 - (b) that evolution occurs.
 - (c) that evolution occurs through a process of gradual change.
 - (d) that the Earth is older than a few thousand years.
 - (e) a mechanism for how evolution occurs.
5. Those being hunted adopt a coloring scheme that allows them to blend in to the colors of the environment.
 - (a) Mimetic coloration

- (b) Aposomatic coloration
- (c) Deceptive markings
- (d) Batesian mimicry
- (e) Flash coloration

True / False (10 marks)

Answer True or False

6. True or False: Is (18)F-FDG a surrogate tracer to measure tumor hypoxia?
7. True or False: PREVALENCE OF THE STREPTOCOCCUS AGALACTIAE IN THE PREGNANT WOMAN FROM THE AUTONOMIC CITY OF MELILLA: IS CULTURE A DETERMINANT FACTOR?
8. True or False: Are the GFRUP's recommendations for withholding or withdrawing treatments in critically ill children applicable?
9. True or False: Do adjuvant aromatase inhibitors increase the cardiovascular risk in postmenopausal women with early breast cancer?
10. True or False: Is dexamethasone an effective alternative to oral prednisone in the treatment of pediatric asthma exacerbations?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the physiological effects of nerve gases that inhibit cholinesterase on the human body, including the mechanisms involved and potential consequences of such exposure.
12. Discuss the mechanisms through which a newborn can contract syphilis, including potential pathways of transmission and the implications for maternal health.

End of Paper